AutoMeasure

Measure the **height and width of objects automatically** using your laptop camera or an uploaded photo. No ruler needed.

Features: photo upload, laptop webcam, live preview, automatic object detection (edges or YOLO), boundary outlines, automatic height and width, reference-scale calibration, unit choice (mm, cm, m, inch, feet), measurement history, and CSV export.

Built with: Python, OpenCV, YOLO (Ultralytics), Flask, HTML, CSS, JavaScript, MySQL.

What you need

- Windows PC or laptop with a webcam
- Python 3.12 (3.12 or 3.13 works; **do not use 3.15**, OpenCV does not support it yet)
- MySQL Server and MySQL Workbench
- VS Code (recommended)
- Chrome or Edge browser

Setup (do this once)

Step 1: Get the project

git clone https://github.com/lumani-lc/automeasure.git
cd automeasure

No Git? Click **Code** → **Download ZIP** on the GitHub page, unzip it, and open the folder in VS Code.

Step 2: Create a virtual environment

py -3.12 -m venv venv
venv\Scripts\activate

Your terminal prompt should now start with `(venv)`.

Step 3: Install the packages

pip install -r requirements.txt

This also installs YOLO and PyTorch, so it can take a few minutes.

Step 4: Create the database

1. Open **MySQL Workbench** and connect to your local server.
2. Choose **File** → **Open SQL Script** and select `schema.sql`.
3. Click the **lightning bolt** to run it.
4. Refresh the Schemas panel. You should see `automeasure` → `measurements`.

Note: The **MySQL Workbench** is it depends on you.

Step 5: Add your MySQL password

1. Copy `config.example.py` and name the copy `config.py`.

Open `config.py` and type your MySQL root password:
DB_CONFIG = dict(host="localhost", user="root", password="YOUR_PASSWORD", database="automeasure")

2.

Run the app

venv\Scripts\activate
python app.py

Open **http://127.0.0.1:5000** in Chrome or Edge. (Use `127.0.0.1` or `localhost`, otherwise the camera is blocked.)

How to use it

1. Click **Start camera**, or **Upload photo**.
2. **Calibrate** first:
    - Put a reference object of known size (for example a credit card, 8.56 cm) as the **left-most** object, on the same surface as the things you will measure.

    - Enter its **longest side**, choose the unit, and click **Calibrate**.
3. Put the object(s) to measure on the surface and click **Measure now**, or turn on **Live measuring**.
4. Green outlines and sizes appear on the objects. Change the **Unit** dropdown any time.
5. Choose the **Detection** mode:
    - **Edges**: works on any object on a plain background.
    - **YOLO**: recognises common objects (bottle, cup, phone, book, laptop...). The first use downloads the model and needs internet.
6. Type a label and click **Save to history**.
7. Use **Export CSV** to download all saved measurements.

Tips for accurate results

- Use a **plain, light surface** and **even lighting** with no strong shadows.
- Keep the camera **straight above** the objects, not at an angle.
- Keep the reference and the objects at the **same height and distance** from the camera.
- **Calibrate again** whenever you move the camera.
- Taller objects look bigger to the camera, so the reading can be slightly high.

Troubleshooting

Problem	Fix
`py -3.12` not found	Install Python 3.12, then reopen the terminal
NumPy / OpenCV error on start	You are on the wrong Python version. Rebuild the venv with 3.12
`Access denied for user 'root'`	Wrong password in `config.py`
`Unknown database 'automeasure'`	Run `schema.sql` in MySQL Workbench
Camera does not start	Allow camera permission, close other apps using the camera, use `127.0.0.1`
No objects detected	Use a plainer background, better light, or try the other Detection mode

"Unexpected token '<'" when saving	The server crashed. Read the last line of the terminal error

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Project files

File	Purpose
`app.py`	Flask server and database routes
`measure.py`	Edge-based object and boundary detection
`yolo_detect.py`	YOLO object detection
`schema.sql`	MySQL database and table
`templates/index.html`	Web page
`static/app.js`, `static/style.css`	Camera, drawing, calibration, history

Limitations

- Measures width and height in the camera plane only (no depth).
- Accuracy depends on lighting, background, camera angle, and a correct reference object.
- YOLO only recognises the 80 object types it was trained on, and it uses straight (not rotated) boxes.